

Generalized Bayes Estimator $\delta_\pi(x)$ w/ an improper prior $\pi(\theta)$ (not a probability measure, but some σ -finite measure), is the estimator minimizing the posterior risk $E[L(\theta, \delta(x)) | X] \forall x \in \mathcal{X}$.

Ex $X_1, \dots, X_n \text{ i.i.d. } N(0, \sigma^2)$, σ^2 known, $\theta \in \Theta = \mathbb{R}$, but $\pi(\theta) = 1 \forall \theta \in \mathbb{R}$.
Then $\pi(\theta)$ is a non-informative, improper prior.

The j.t. as always, is $f(x, \theta) = f(x | \theta) \pi(\theta)$

$$= \frac{1}{(\sigma\sqrt{2\pi})^n} \exp. -\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \theta)^2$$

If the ~~pdf~~ marginal $f_X(x) = \int f(x, \theta) d\theta$ is finite (and non-zero),
 $\pi(\theta | x) = \frac{f(x, \theta) \pi(\theta)}{f_X(x)}$ will ^(P) be a proper posterior distn, in which case

the generalized Bayes estimator is δ_{GB} if the posterior risk $E[\text{sq. error loss} | X]$ is minimized at $\delta_{GB} \forall x \in \mathcal{X}$.

If $\pi(\theta | x)$ is not a proper prob. distn, then the above does not work and Bayes estimator (w/ any loss fn) cannot be obtained in the conventional way

Here, $f_X(x) = \frac{1}{(\sigma\sqrt{2\pi})^n} \int_{-\infty}^{\infty} \exp. -\frac{1}{2\sigma^2} \sum (x_i - \theta)^2 d\theta = \frac{1}{(\sigma\sqrt{2\pi})^n} \exp. -\frac{(n-1)s^2}{2\sigma^2} \cdot \frac{\sqrt{2\pi}\sigma}{\sqrt{n}}$,
where, $s^2 = \frac{1}{n-1} \sum (x_i - \bar{x})^2$

$\Rightarrow \pi(\theta | x) = \frac{\sqrt{n}}{\sqrt{2\pi}\sigma} \exp. -\frac{n}{2\sigma^2} (\bar{x} - \theta)^2 \equiv N(\bar{x}, \sigma^2/n)$ is in fact a prob. distn

W/ the squared error loss fn, the generalized Bayes estimator of θ is

$\delta_{GB} = \bar{X}$, which is in fact, the UMVUE of θ .

[Recall, the result in Lecture 37, that w/ the sq. error loss, no unb. est. $\delta(x)$ of $\theta(\theta)$ can be its Bayes estimator (except in degenerate case) ~~is~~ irrespective of any prior, should now be modified as 'irrespective of any proper prior']

Why do I need the prior to be a prob. distn to work out this result?

For any $\alpha \in (0,1)$, a $100(1-\alpha)\%$ credible set for θ is a set $C \ni$

$$P_{\theta|x}(\theta \in C(x)) \geq \alpha \quad ('>' \text{ requirement is for discrete posterior distns})$$

$$\Leftrightarrow \int_{C(x)} \pi(\theta|x) d\theta \geq \alpha.$$

For any $\alpha \in (0,1)$, a ~~100~~ $100(1-\alpha)\%$ highest posterior density (HPD) credible set for θ is the set $C(x) = \{\theta; \pi(\theta|x) \geq k\}$, where k is $\ni \int_{C(x)} \pi(\theta|x) d\theta \geq 1-\alpha$.

Note (i) The credible set is called HPD credible set as it contains values of parameters for which the posterior density is maximum (i.e., highest).

(ii) If the posterior density is unimodal, then the ^{$100(1-\alpha)\%$} HPD credible set is the interval with shortest length among all $100(1-\alpha)\%$ credible intervals.

Ex 1 $X_1, \dots, X_n$ iid $N(\theta, \sigma^2)$, σ^2 known, $\theta \in \mathbb{R}$

$\theta \sim N(\mu_0, \sigma_0^2)$, so that $\theta|x \sim N(\eta, \lambda^2)$ where $\eta = \frac{\frac{n\bar{x}}{\sigma^2} + \frac{\mu_0}{\sigma_0^2}}{\frac{n}{\sigma^2} + \frac{1}{\sigma_0^2}}$, $\lambda^2 = \frac{1}{\frac{n}{\sigma^2} + \frac{1}{\sigma_0^2}}$.

$$\therefore P_{\theta|x} \left(\frac{|\theta - \eta|}{\lambda} \leq t_{\alpha/2} \right) = 1 - \alpha \quad (\text{meets the } 100(1-\alpha)\% \text{ criterion})$$

$\therefore$ a natural choice of the cut-off pts $\pm t_{\alpha/2}$ is taken, which also ensures an equal tailed ^{$100(1-\alpha)\%$} credible interval, viz, $\eta \pm t_{\alpha/2} \lambda$ for θ .

Case ① $\eta=1, \mu_0=0, \sigma^2=1$, a $100(1-\alpha)\%$ credible interval is $\frac{\bar{x}}{1 + \frac{1}{\sigma_0^2}} \pm t_{\alpha/2} \frac{1}{\sqrt{1 + \frac{1}{\sigma_0^2}}}$
Further, if $\sigma_0^2 \rightarrow \infty$, a cr. int. is $\bar{x} \pm t_{\alpha/2}$ (the usual $100(1-\alpha)\%$ confidence interval for $n \geq 1, \sigma^2 \geq 1$, i.e., prior has become useless)

② $\mu_0=0, \sigma^2=\sigma_0^2=1$

here $100(1-\alpha)\%$ credible interval is $\frac{n\bar{x}}{n+1} \pm t_{\alpha/2} \frac{1}{\sqrt{n+1}}$ with length $2t_{\alpha/2} \frac{1}{\sqrt{n+1}}$

Meas, $100(1-\alpha)\%$ confidence interval is $\bar{x} \pm t_{\alpha/2} \frac{1}{\sqrt{n}}$ with length $= 2t_{\alpha/2} \frac{1}{\sqrt{n}}$.

Shorter length because of using extra info through the prior (informative, proper).

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9/38

$x_1, \dots, x_n$ iid $N(\theta, \sigma^2)$, $\pi(\theta) \propto 1$, $\theta \in \mathbb{R}$
prior π is non-informative, improper.

$$j\theta \quad f(\bar{x}; \theta) = \text{const. exp} - \frac{(n-1)s^2}{2\sigma^2} \text{exp.} - \frac{n}{2\sigma^2} (\bar{x} - \theta)^2 \\ = \text{const.}^* \text{exp.} - \frac{n}{2\sigma^2} (\bar{x} - \theta)^2$$

marginal
$$f_X(\bar{x}) = \text{const.}^* \sqrt{2\pi} \frac{\sigma}{\sqrt{n}}$$

hence, $\pi(\theta|\bar{x}) \propto N(\bar{x}, \sigma^2/n)$.

$100(1-\alpha)\%$ credible interval for θ is $\bar{x} \pm t_{\alpha/2} \frac{\sigma}{\sqrt{n}}$, the usual $100(1-\alpha)\%$ confidence int., the choice of prior is not helping to give any extra info'.

d) Computation of HPD credible interval

~~$x_1, \dots, x_n$~~ $x_1, \dots, x_n$ iid $N(\theta, \sigma^2)$, $\sigma^2 > 0$, known, $\theta \in \mathbb{R}$.

$\theta \sim N(\mu_0, \sigma_0^2)$, then $\theta|x \sim N(\eta, \lambda^2)$, $\eta = \frac{\frac{n\bar{x}}{\sigma^2} + \frac{\mu_0}{\sigma_0^2}}{\frac{n}{\sigma^2} + \frac{1}{\sigma_0^2}}$, $\lambda^2 = \frac{1}{\frac{n}{\sigma^2} + \frac{1}{\sigma_0^2}}$.

then
$$C(x) = \{\theta : \pi(\theta|x) \geq k\}.$$

$$= \{\theta : -\frac{1}{2\lambda^2} (\theta - \eta)^2 \geq \ln(k\lambda\sqrt{2\pi})\}$$

$$= \{\theta : |\theta - \eta| \leq \lambda\sqrt{2\pi} (-\ln(k\lambda\sqrt{2\pi}))^{1/2}\}$$

$k \propto \int_{C(x)} \pi(\theta|x) d\theta \geq \alpha$, i.e., $P\left[\frac{|\theta - \eta|}{\lambda} \leq \sqrt{2\pi} (-\ln(k\lambda\sqrt{2\pi}))^{1/2}\right] = 1 - \alpha$

$\Rightarrow k$ can be solved as, $\sqrt{2} (-\ln(k\lambda\sqrt{2\pi}))^{1/2} = t_{\alpha/2}$

$\Rightarrow k = \frac{1}{\lambda\sqrt{2\pi}} e^{-\frac{1}{2} t_{\alpha/2}^2}$, so that

the $100(1-\alpha)\%$ HPD credible interval coincides with the equal-tailed

$100(1-\alpha)\%$ credible interval discussed at the beginning

of this example (before the 4 different cases).